

HW 4.2.1.b: Solving Trigonometric Equations

Solve the trigonometric equation using the unit circle. Answers should be $0 \leq \theta \leq 2\pi$. Each question may have multiple answers.

1.) $\sin \theta = \frac{\sqrt{2}}{2}$

2.) $\cos \theta = -\frac{\sqrt{3}}{2}$

3.) $\tan \theta = -1$

4.) $\sin \theta = -\frac{\sqrt{3}}{2}$

5.) $\cos \theta = 0$

6.) $\csc \theta = \sqrt{2}$

7.) $\sec \theta = \frac{2\sqrt{3}}{3}$

8.) $\sin \theta = -1$

9.) $\tan \theta = \sqrt{3}$

10.) $\cos \theta = \frac{1}{2}$

11.) $\tan \theta = \text{undefined}$

12.) $\sin \theta = 1$

13.) $\cos \theta = \frac{\sqrt{3}}{2}$

14.) $\cot \theta = -\sqrt{3}$

15.) $\csc \theta = -2$

16.) $\sec \theta = -1$

17.) $\tan \theta = \frac{\sqrt{3}}{3}$

18.) $\sin \theta = -\frac{1}{2}$

19.) $\sec \theta = \text{undefined}$

20.) $\cos \theta = 1$

21.) $\csc \theta = -\frac{2\sqrt{3}}{3}$

22.) $\sin \theta = \frac{\sqrt{3}}{2}$

23.) $\cot \theta = -\frac{\sqrt{3}}{3}$

24.) $\cos \theta = -\frac{\sqrt{2}}{2}$